FILTER DESIGN TEST CASE

FBAR (Film Bulk Acoustic Resonator) - Band 12 (700 MHz) for Automotive V2X Communication
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a FBAR (Film Bulk Acoustic Resonator) filter targeting Band 12 (700 MHz) for Automotive V2X Communication. The filter is designed on Langasite substrate with a target center frequency of 3147.0 MHz and bandwidth of 105.0 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Automotive V2X Communication module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: FBAR (Film Bulk Acoustic Resonator)
- Frequency Band: Band 12 (700 MHz)
- Application: Automotive V2X Communication
- Substrate: Langasite
- Package: SiP (System in Package)
- Design Tool: Cadence AWR Microwave Office
- Design Center: Andover HQ Lab

This specification applies to all variants of the FBAR (Film Bulk Acoustic Resonator) design targeting Band 12 (700 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency: 3147.0 MHz
Bandwidth (3 dB): 105.0 MHz
Insertion Loss Target: <= 1.5 dB
Return Loss Target: >= 14.0 dB
Out-of-Band Rejection: >= 26.0 dB
Isolation (Duplexer): >= 43.0 dB
Q Factor: 13694
Temperature Coefficient:-23.8 ppm/°C
Die Size: 1.8 x 2.3 mm
Wafer Size: 6-inch
Number of Resonators: 6
Electrode Material: Mo

Passivation: SiO2/Si3N4 stack

4. TEST PROCEDURE

1. Open Cadence AWR Microwave Office and load the FBAR (Film Bulk Acoustic Resonator) template project.
2. Define the substrate stack: Langasite with specified layer thicknesses.
3. Set design targets: center freq 3147.0 MHz, BW 105.0 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.5 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (SiP (System in Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, Langasite).
10. Perform on-wafer probing using Keysight N9041B UXA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 9441 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.5 dB
- Return loss in passband shall be >= 14.0 dB
- Out-of-band rejection at +/- 158 MHz offset >= 26.0 dB
- Group delay variation across passband shall be <= 5 ns
- Temperature drift over -40°C to +85°C shall be <= 9.4 MHz
- Power handling shall be >= +29 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 98%

6. TEST RESULTS

Measurement Date: 2024-04-22
Devices Tested: 34
Insertion Loss (meas): 1.23 dB
Return Loss (meas): 16.2 dB
Rejection (meas): 28.6 dB
Isolation (meas): 48.2 dB
Q Factor (meas): 13694
Group Delay Variation: 3.4 ns
Power Handling: +31 dBm
Die Yield: 92.3%

Result Classification: FAIL

7. OBSERVATIONS AND FINDINGS

a) The FBAR (Film Bulk Acoustic Resonator) design demonstrated below-target performance for Band 12 (700 MHz).
b) Insertion loss met target specification.
c) Rejection at near-band offset requires additional resonator tuning.
d) Simulation-to-measurement correlation was within 3% for all key parameters.
e) Batch-to-batch reproducibility confirmed over 3 wafer lots.

8. CORRECTIVE ACTIONS

1. Optimize resonator geometry to reduce insertion loss by ~0.3 dB.
2. Re-run EM simulation with refined mesh for better accuracy.
3. Escalate to advanced materials team for alternative piezoelectric stack.

9. SIGN-OFF

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